EXERCISE SET 1

EXERCISE 1

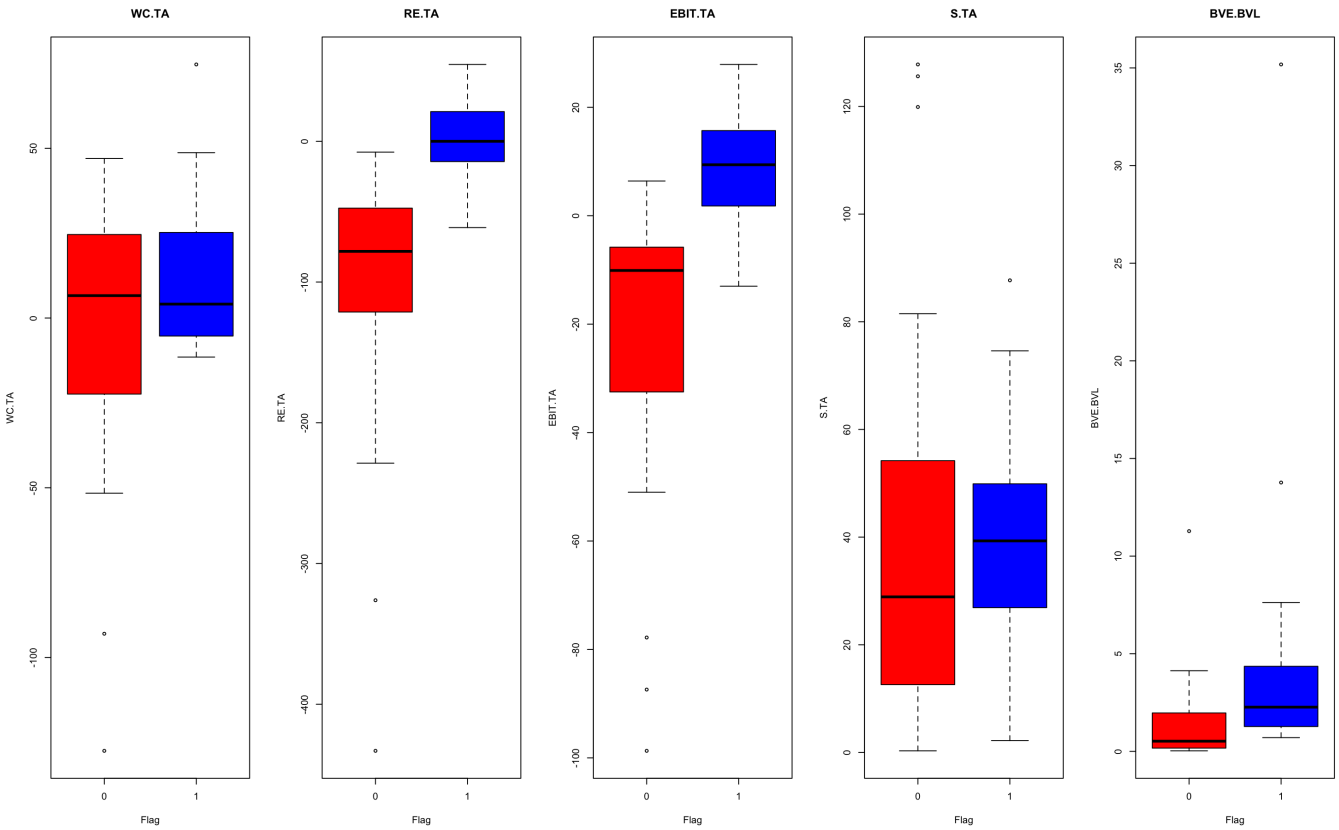
```
dati <- read.table(choose.files(), header=TRUE)
summary(dati)
dati$Flag<-1-dati$Bankrupt #we consider "good" the companies that are not
bankrupt
```

a) Present the box plots of all the 5 covariates and comment on which ones seem to have a good discriminatory power.

```
color1<-c("red","blue") #we create a vector of colors that represent bad and
good units

par(mfrow=c(1,5)) #in this way we are able to see the boxplot for each
variable in one graph

boxplot(dati$WC.TA~dati$Flag,xlab = "Flag",ylab = "WC.TA",col=color1)
boxplot(dati$RE.TA~dati$Flag,xlab = "Flag",ylab = "RE.TA",col=color1)
boxplot(dati$EBIT.TA~dati$Flag,xlab = "Flag",ylab = "EBIT.TA",col=color1)
boxplot(dati$S.TA~dati$Flag,xlab = "Flag",ylab = "S.TA",col=color1)
boxplot(dati$BVE.BVL~dati$Flag,xlab = "Flag",ylab = "BVE.BVL",col=color1)
```



To determine which variables have good discriminatory power, we looked at how well the boxplots separate the two groups (0 and 1). In fact, a good discriminator should show minimal overlap between the distributions of the two categories.
At first glance, we can notice that RE.TA and EBIT.TA separate very well the two populations but it is interesting to see what could change if we transform the variables. It will help us also to get an easier interpretation of the graphs.

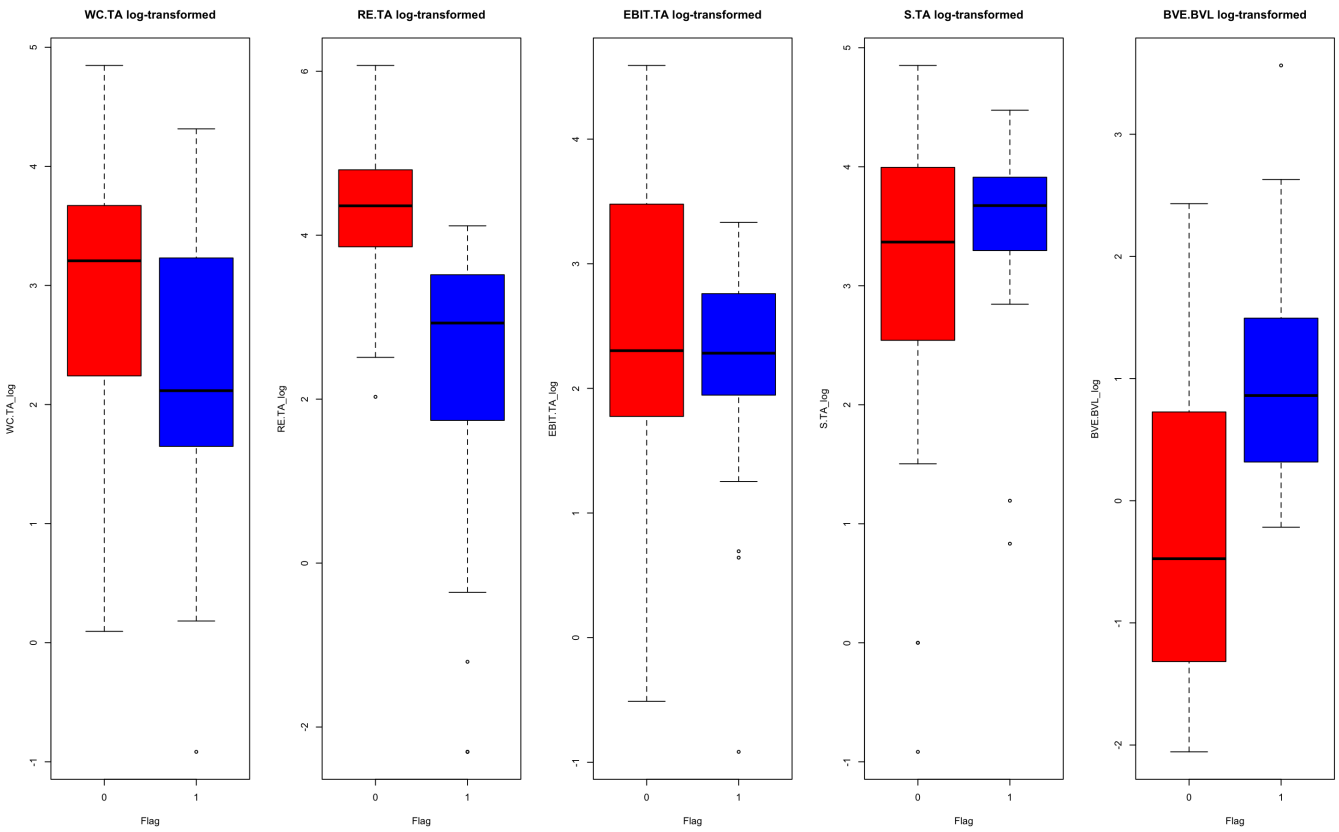
```
min(dati$WC.TA); min(dati$RE.TA); min(dati$EBIT.TA); min(dati$S.TA);
min(dati$BVE.BVL)
[1] -127.5
[1] -433.1
[1] -98.7
[1] 0.3
[1] 0.028
```

There is an issue with applying the logarithm to the variables because WC.TA, RE.TA, and EBIT.TA are negative. Since the logarithm can only be applied to positive values, this poses a problem. This characteristic can also be observed by examining their minimum values.

To avoid this problem, we can take the absolute value of the negative variables before computing the logarithm.
This way, the negative values will be shifted to the positive range while still preserving their

original sign distinction (Flag = 0 or Flag = 1).
This process is useful for ensuring comparability among all variables after the transformation.

```
#creation of the variables for the log-transformation
WC.TA_log<-log(abs(dati$WC.TA+0.1))
RE.TA_log<-log(abs(dati$RE.TA+0.1))
EBIT.TA_log<-log(abs(dati$EBIT.TA+0.1))
S.TA_log<-log(dati$S.TA+0.1)
BVE.BVL_log<-log(dati$BVE.BVL+0.1)
#now we are able to construct the boxplots
boxplot(WC.TA_log~dati$Flag,xlab = "Flag",ylab = "WC.TA_log",col=color1,main
= "WC.TA log-transformed")
boxplot(RE.TA_log~dati$Flag,xlab = "Flag",ylab = "RE.TA_log",col=color1,main
= "RE.TA log-transformed")
boxplot(EBIT.TA_log~dati$Flag,xlab = "Flag",ylab =
"EBIT.TA_log",col=color1,main = "EBIT.TA log-transformed")
boxplot(S.TA_log~dati$Flag,xlab = "Flag",ylab = "S.TA_log",col=color1,main =
"S.TA log-transformed")
boxplot(BVE.BVL_log~dati$Flag,xlab = "Flag",ylab = "BVE.BVL_log",col=color1,
main = "BVE.BVL log-transformed")
```

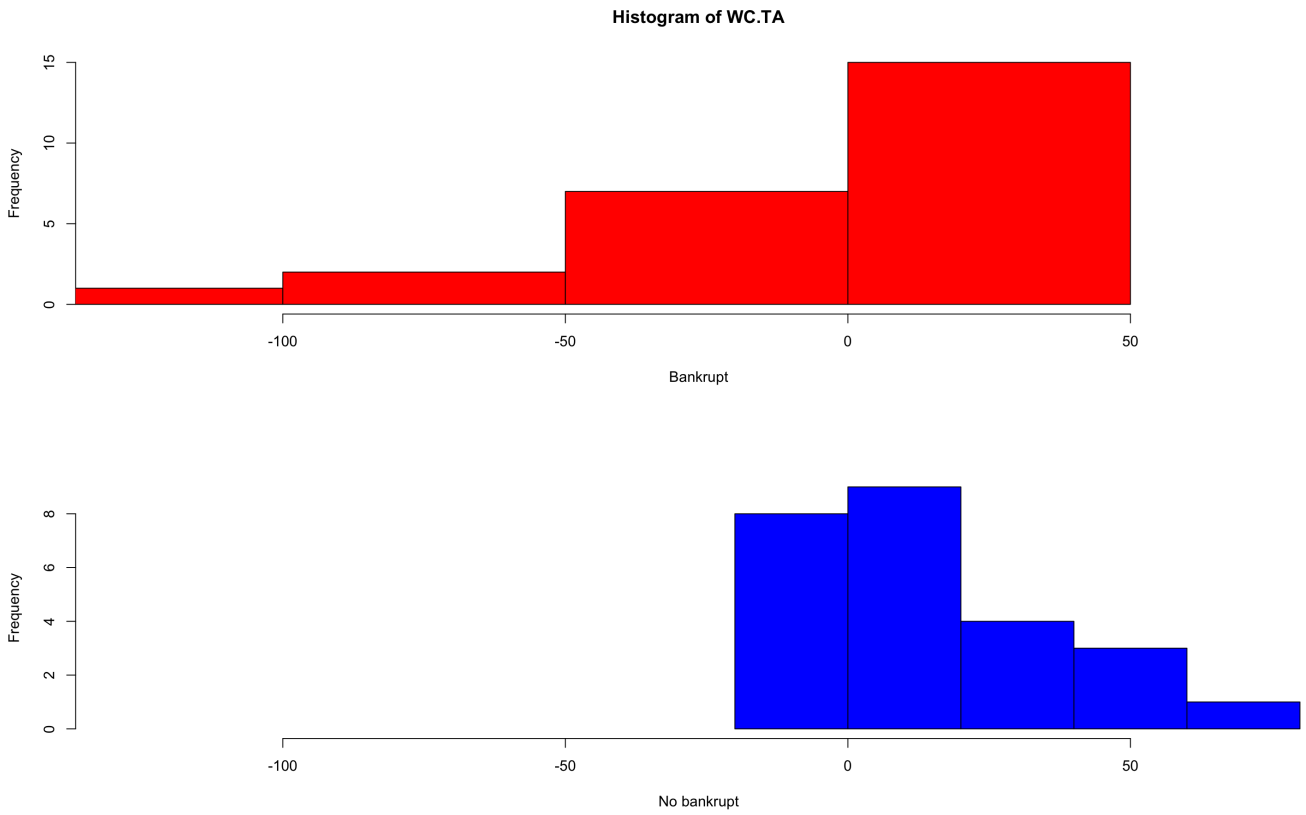


The variable with the best discriminative power appears to be RE.TA. In the RE.TA plot, there is a clearer separation between the two distributions: the distribution with Flag=0 is more concentrated at the top, while the distribution with Flag=1 is lower and less overlapping.

Other variables like S.TA and EBIT.TA show some difference between the distributions, but with a less distinct separation compared to RE.TA. The variables WC.TA and BVE.BVL seem to have more limited discriminative power, with more overlapping distributions. RE.TA appears to be the variable with the best discriminative power to distinguish between the two groups represented by the flag.

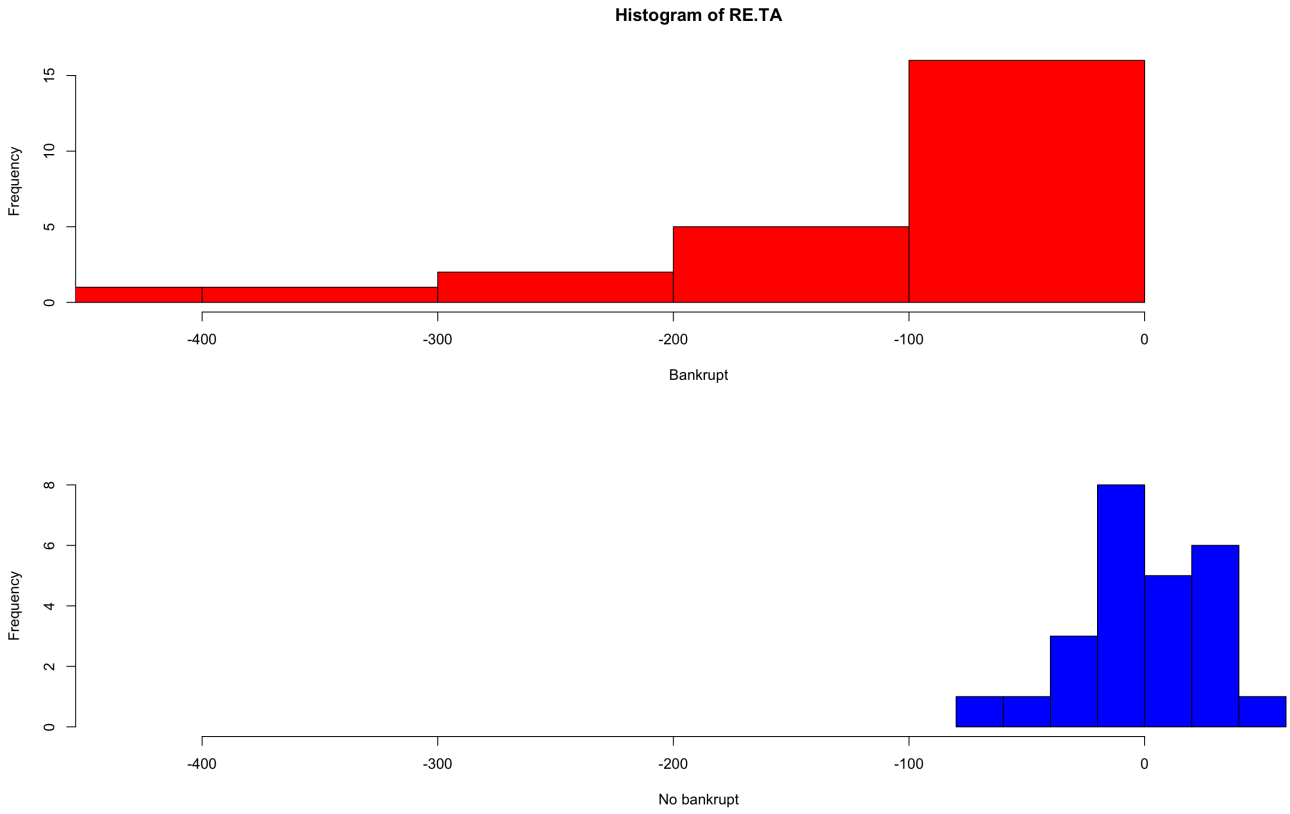
Then present the histograms of the distribution of good firms and bad firms according to each indicator. Evaluate if a log transform or other transform can bring the distribution closer to normality.

```
par(mfrow=c(2,1))
#WC.TA
#bad units
hist(dati$WC.TA[dati$Flag==0], xlim=range(min(dati$WC.TA)-1,
max(dati$WC.TA)+1),
     xlab = "Bankrupt",main = "Histogram of WC.TA",col = "red")
#good unts
hist(dati$WC.TA[dati$Flag==1], xlim=range(min(dati$WC.TA)-1,
max(dati$WC.TA)+1),
     xlab = "No bankrupt",main = "", col = "blue")
```

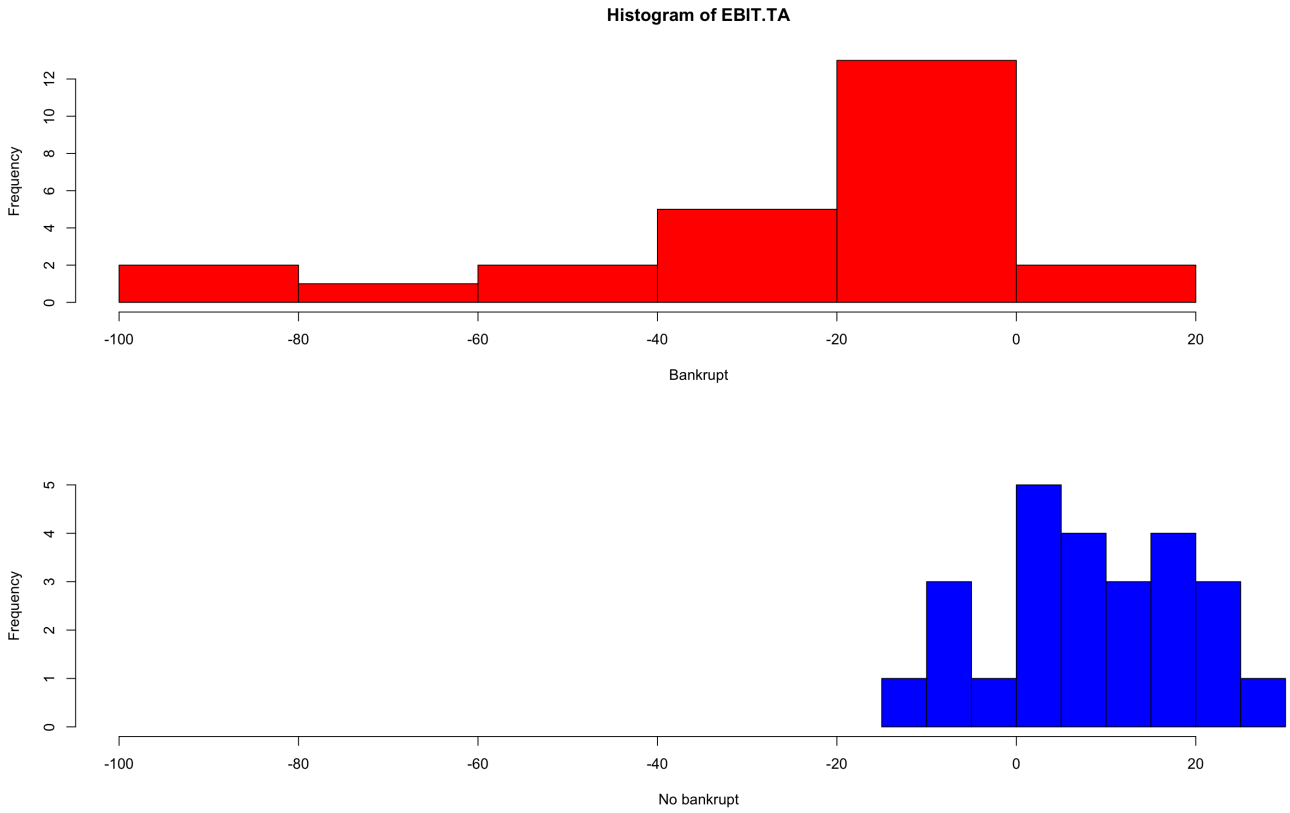



```
#RE.TA
#bad units
```

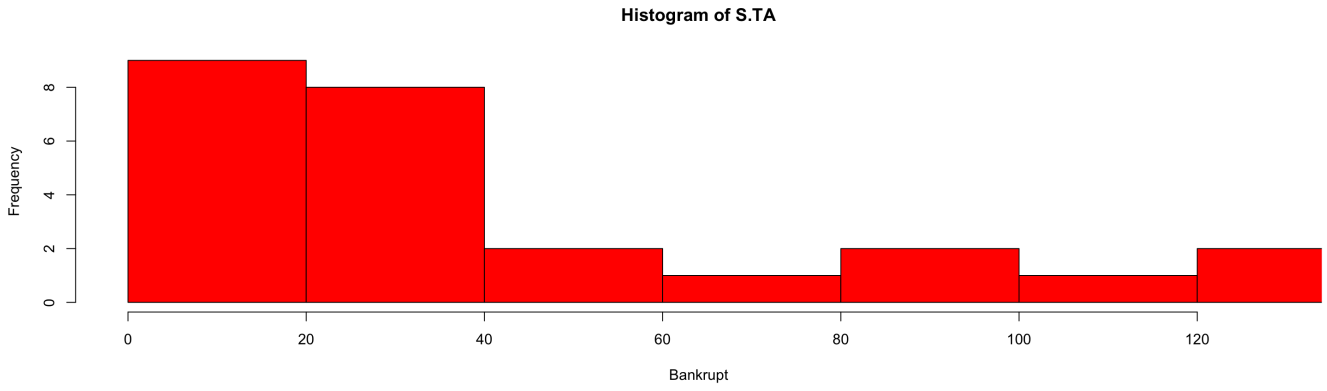
```
hist(dati$RE.TA[dati$Flag==0], xlim=range(min(dati$RE.TA)-1,
max(dati$RE.TA)+1),xlab = "Bankrupt",main = "Histogram of RE.TA",col =
"red")
#good units
hist(dati$RE.TA[dati$Flag==1], xlim=range(min(dati$RE.TA)-1,
max(dati$RE.TA)+1),xlab = "No bankrupt",main = "",col = "blue")
```

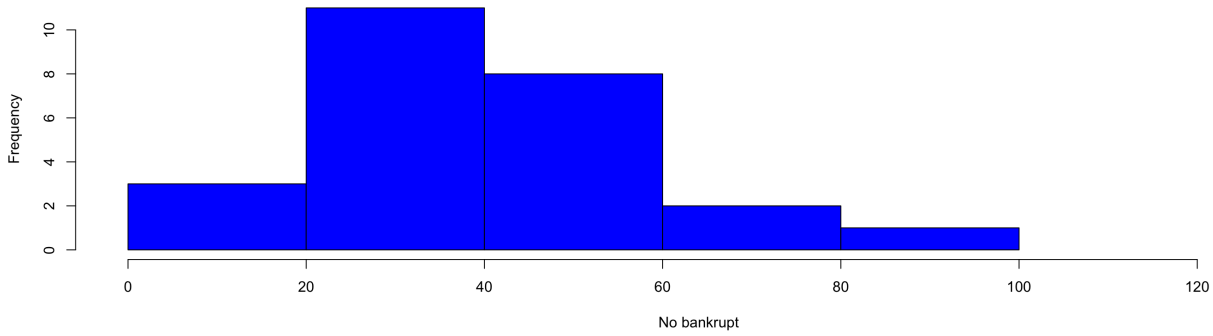



```
#EBIT.TA
#bad units
hist(dati$EBIT.TA[dati$Flag==0],
xlim=range(min(dati$EBIT.TA)-1,max(dati$EBIT.TA)+1),xlab = "Bankrupt",main =
"Histogram of EBIT.TA", col = "red")
#good units
hist(dati$EBIT.TA[dati$Flag==1],
xlim=range(min(dati$EBIT.TA)-1,max(dati$EBIT.TA)+1),xlab = "No
bankrupt",main = "", col = "blue")
```

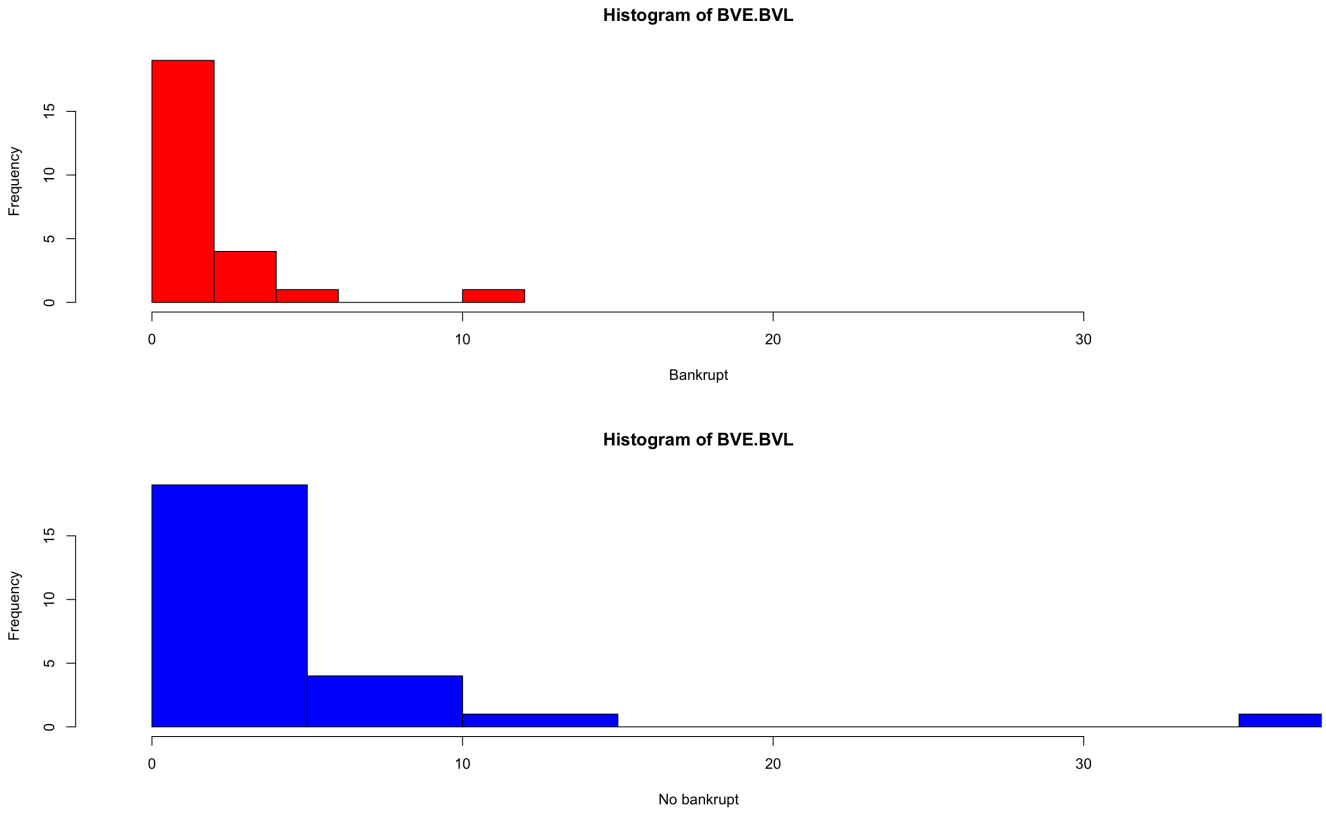



```r
#S.TA
#bad units
hist(dati$S.TA[dati$Flag==0], xlim=range(min(dati$S.TA)-1,
max(dati$S.TA)+1),xlab = "Bankrupt",main = "Histogram of S.TA",col = "red")
#good units
hist(dati$S.TA[dati$Flag==1], xlim=range(min(dati$S.TA)-1,
max(dati$S.TA)+1),xlab = "No bankrupt",main = "",col = "blue")
```



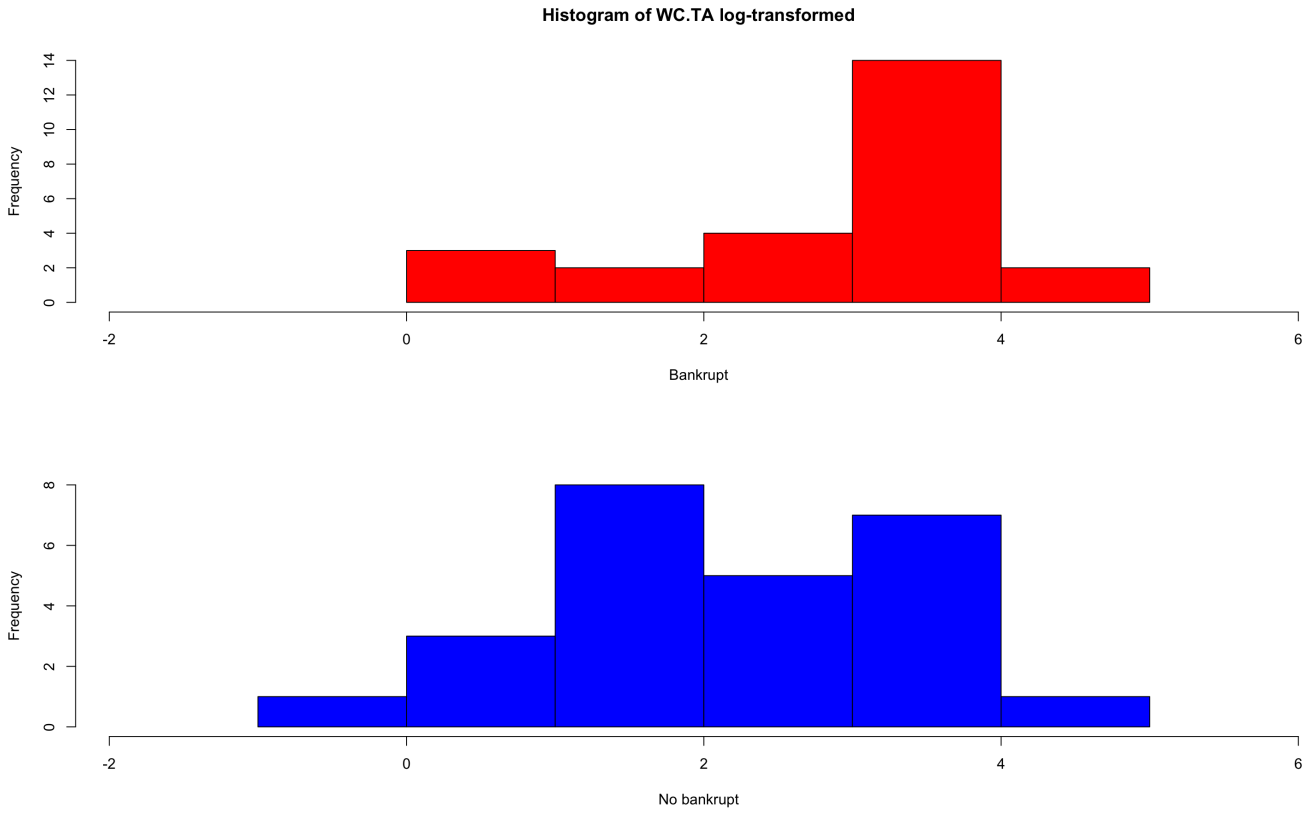



```r
#BVE.BVL
#bad units
hist(dati$BVE.BVL[dati$Flag==0], xlim=range(min(dati$BVE.BVL)-1,
max(dati$BVE.BVL)+1),xlab = "Bankrupt",main = "Histogram of BVE.BVL",col =
"red")
#good units
hist(dati$BVE.BVL[dati$Flag==1],
xlim=range(min(dati$BVE.BVL)-1,max(dati$BVE.BVL)+1),xlab = "No
bankrupt",main = "",col = "blue")
```

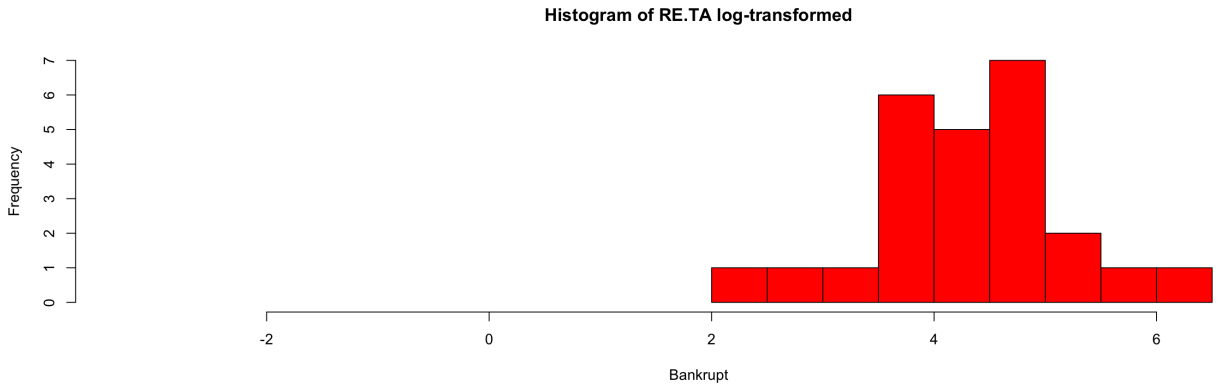


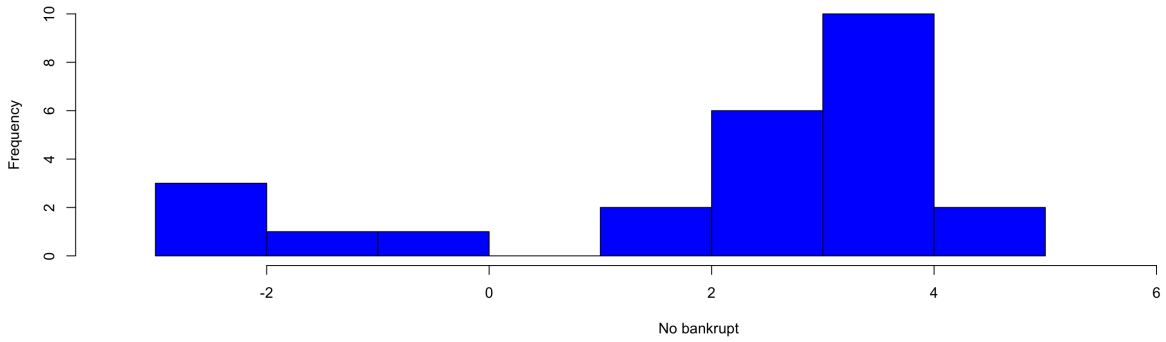
The logarithm transformation is the following:

```
par(mfrow=c(2,1))
#WC.TA_log
#bad units
hist(WC.TA_log[dati$Flag==0], xlim=range(min(WC.TA_log)-1,max(WC.TA_log)+1),
     xlab = "Bankrupt",main = "Histogram of WC.TA log-transformed",col =
"red")
#good units
hist(WC.TA_log[dati$Flag==1], xlim=range(min(WC.TA_log)-1,
max(WC.TA_log)+1),
     xlab = "No bankrupt",main = "", col = "blue")
```

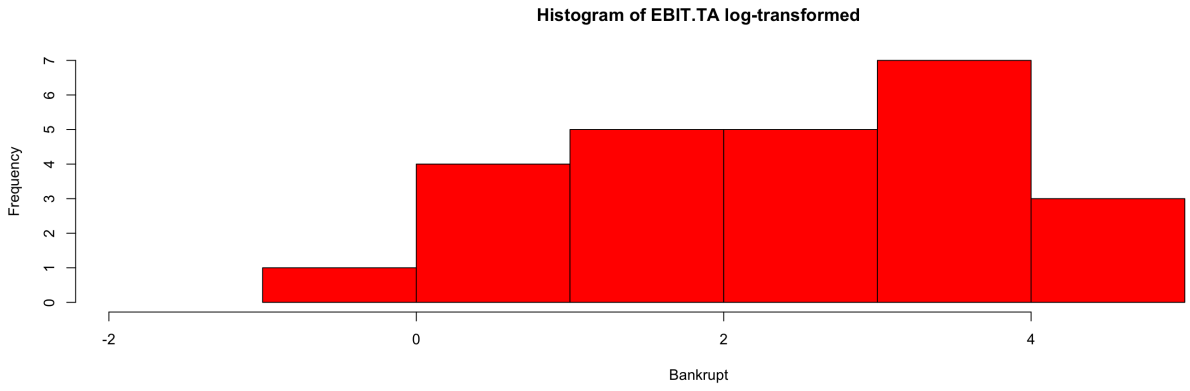


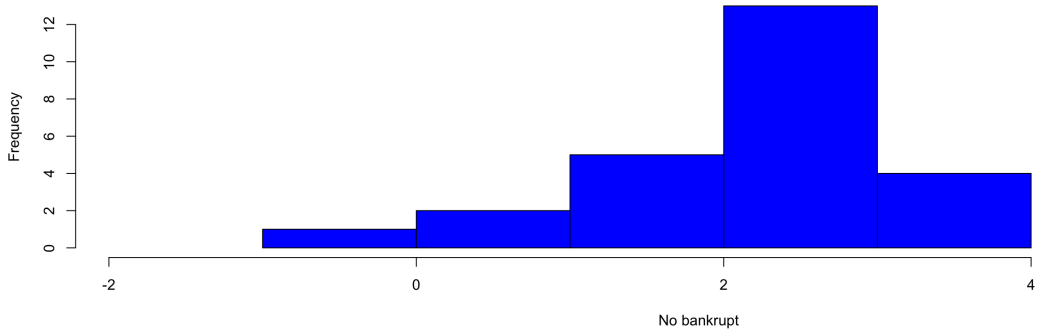

```
#RE.TA_log
#bad units
hist(RE.TA_log[dati$Flag==0], xlim=range(min(RE.TA_log)-1,
max(RE.TA_log)+1),
     xlab = "Bankrupt",main = "Histogram of RE.TA log-transformed",col =
"red")
#good units
hist(RE.TA_log[dati$Flag==1], xlim=range(min(RE.TA_log)-1,
max(RE.TA_log)+1),
     xlab = "No bankrupt",main = "",col = "blue")
```



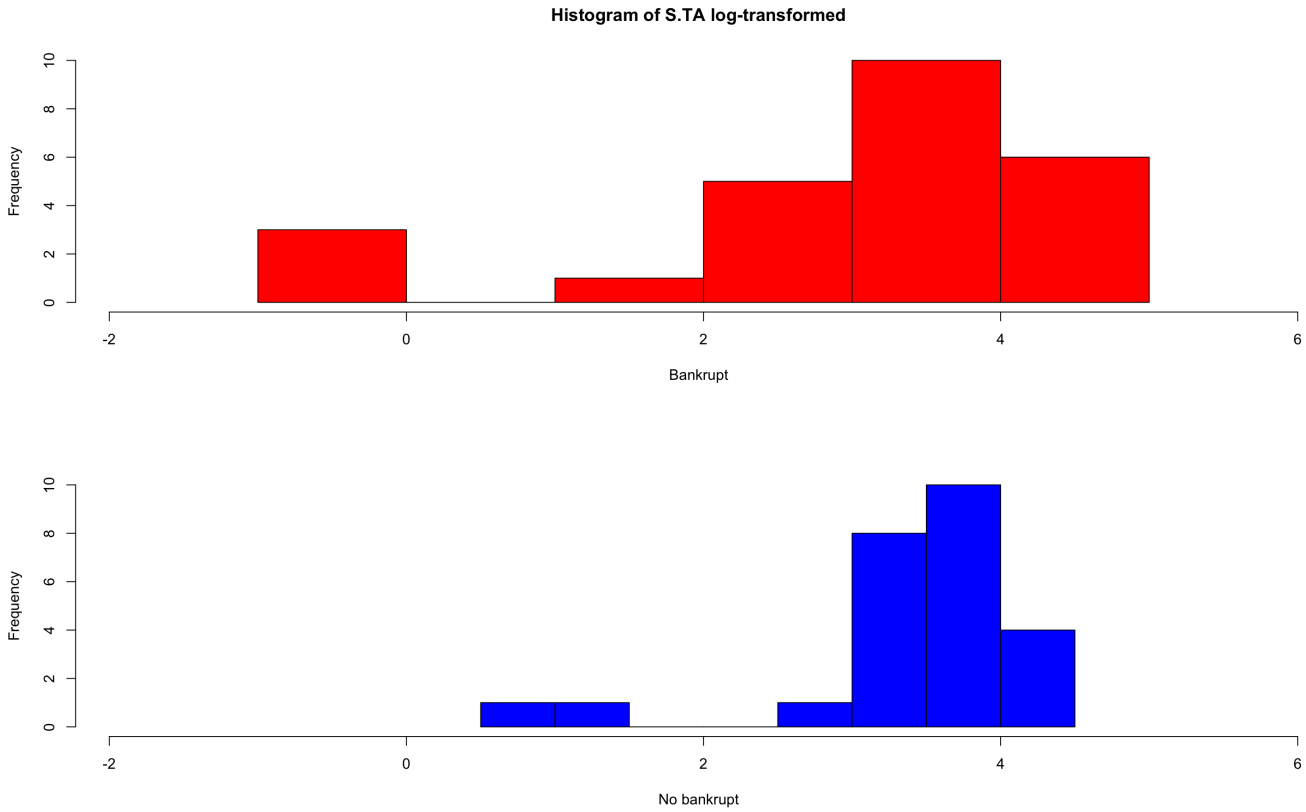



```
#EBIT.TA_log
#bad units
hist(EBIT.TA_log[dati$Flag==0], xlim=range(min(EBIT.TA_log)-1,
max(EBIT.TA_log)+1),
     xlab = "Bankrupt",main = "Histogram of EBIT.TA log-transformed", col =
"red")
#good units
hist(EBIT.TA_log[dati$Flag==1],
xlim=range(min(EBIT.TA_log)-1,max(EBIT.TA_log)+1),
     xlab = "No bankrupt",main = "", col = "blue")
```

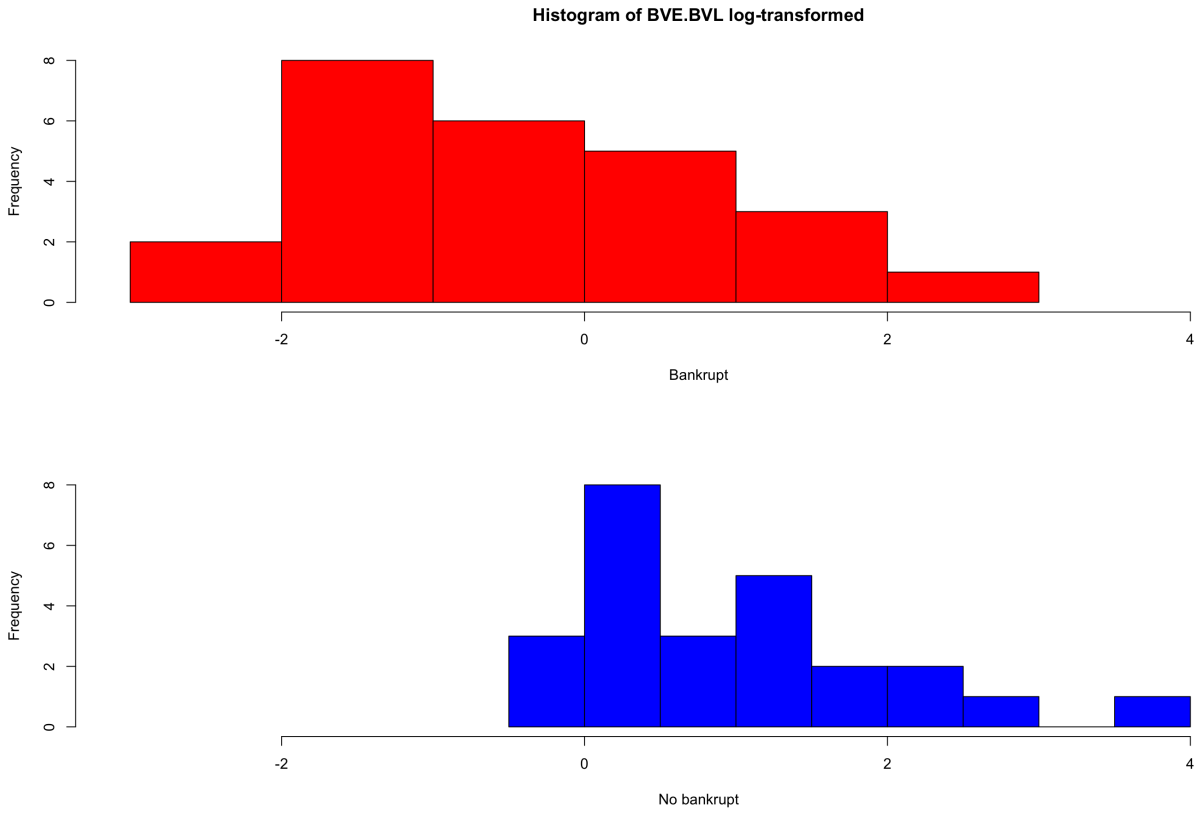





```
#S.TA_log
#bad units
hist(S.TA_log[dati$Flag==0], xlim=range(min(S.TA_log)-1, max(S.TA_log)+1),
     xlab = "Bankrupt",main = "Histogram of S.TA log-transformed",col =
"red")
#good units
hist(S.TA_log[dati$Flag==1], xlim=range(min(S.TA_log)-1, max(S.TA_log)+1),
     xlab = "No bankrupt",main = "",col = "blue")
```

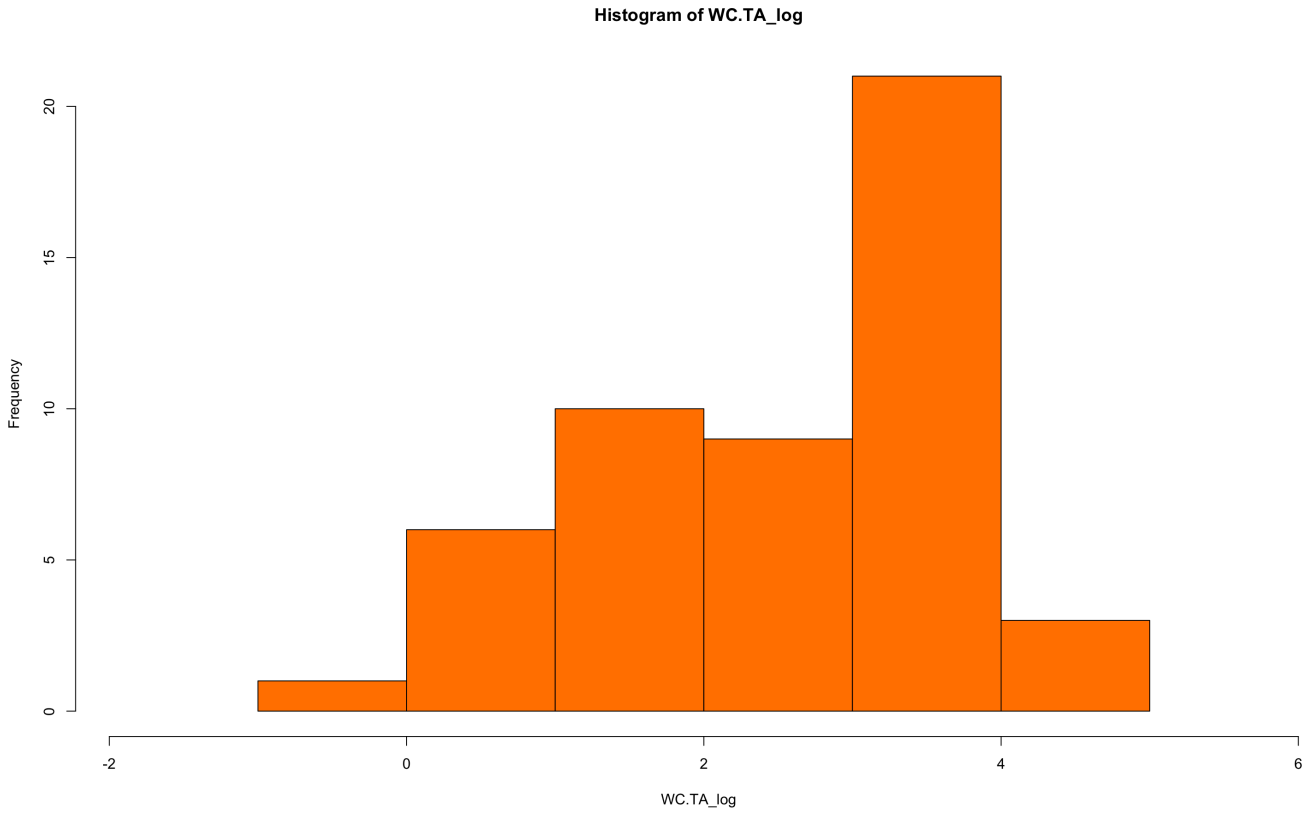



```
#BVE.BVL_log
#bad units
hist(BVE.BVL_log[dati$Flag==0],
xlim=range(min(BVE.BVL_log)-1,max(BVE.BVL_log)+1),
     xlab = "Bankrupt",main = "Histogram of BVE.BVL log-transformed",col =
"red")
#good units
hist(BVE.BVL_log[dati$Flag==1],
xlim=range(min(BVE.BVL_log)-1,max(BVE.BVL_log)+1),
     xlab = "No bankrupt",main = "",col = "blue")
```

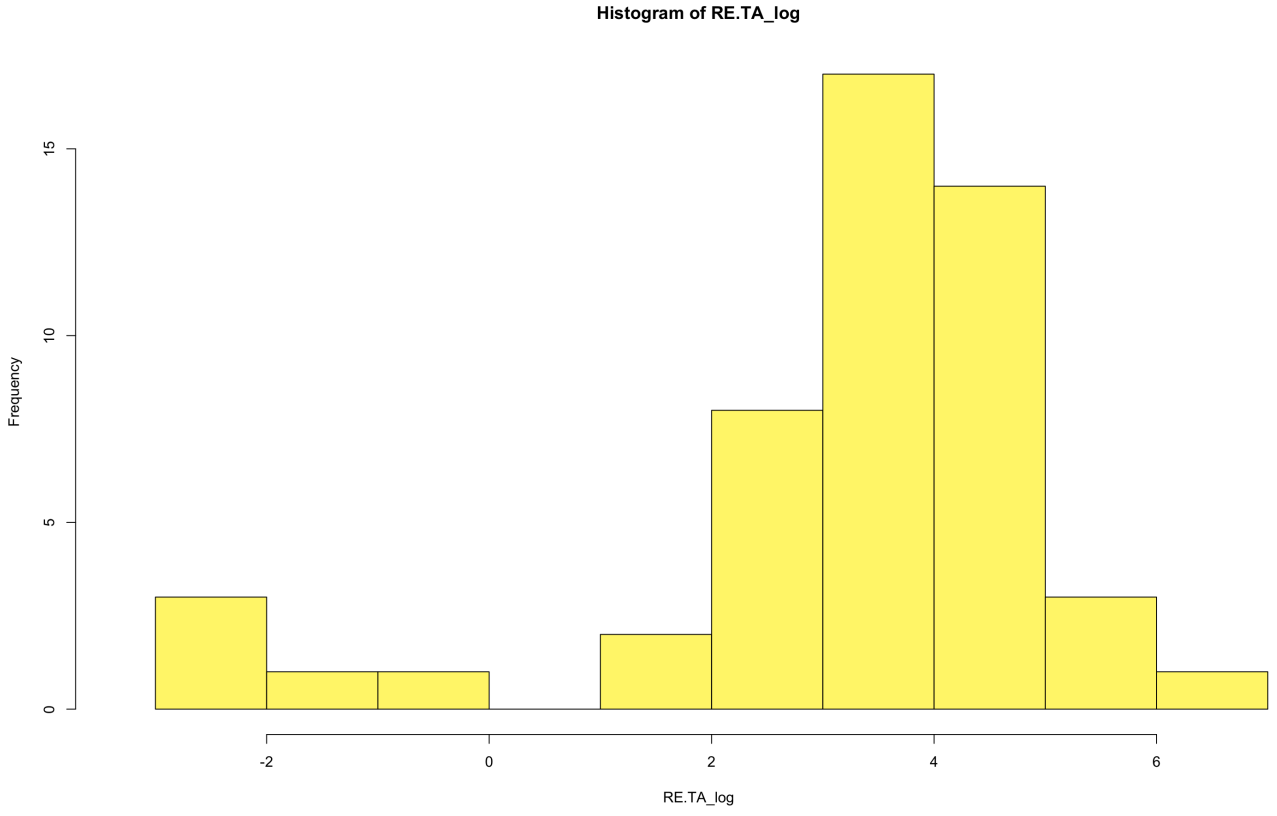


Once the transformations are applied to all the variables, we can check their normality.

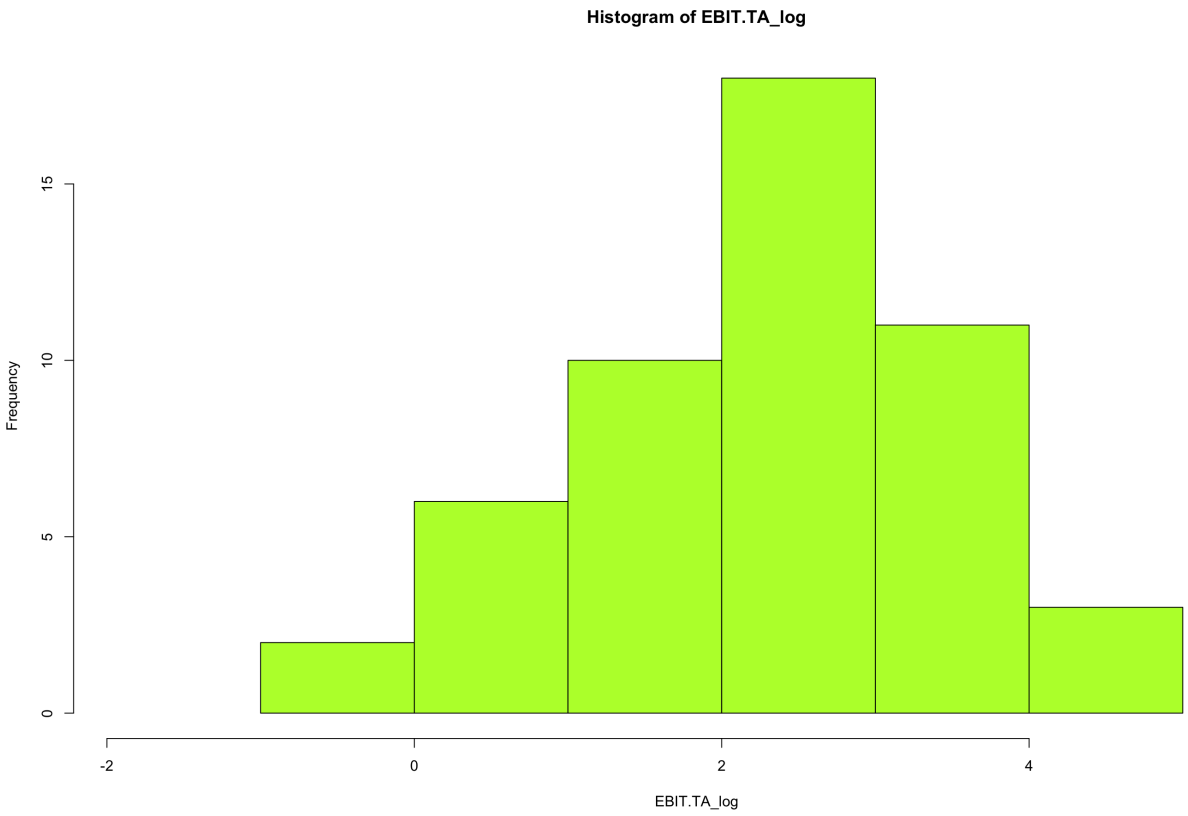
```
par(mfrow=c(1,1))
#WC.TA_log
hist(WC.TA_log, xlim=range(min(WC.TA_log)-1, max(WC.TA_log)+1),
     xlab = "WC.TA_log", main = "Histogram of WC.TA_log", col =
"darkorange")
```



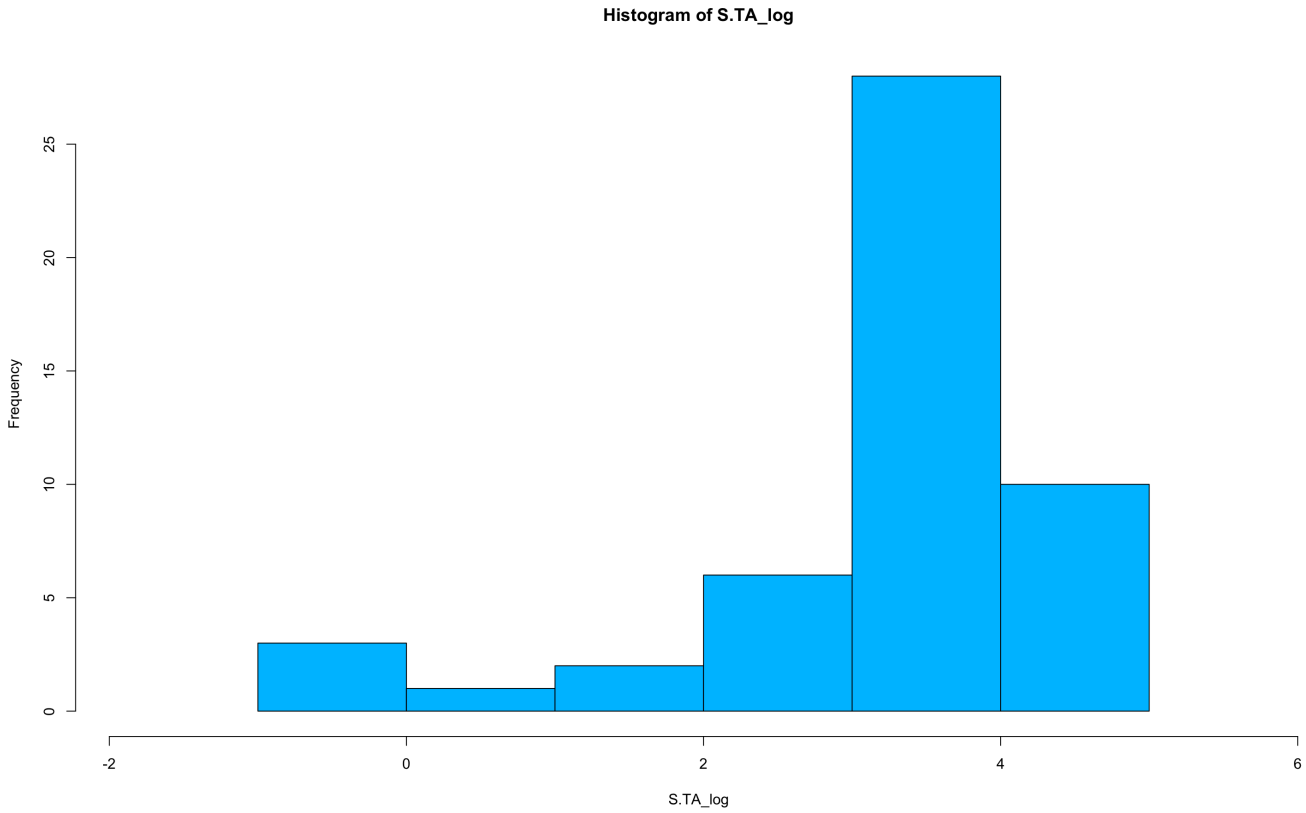

```r
#RE.TA_log
hist(RE.TA_log, xlim=range(min(RE.TA_log)-1, max(RE.TA_log)+1),xlab =
"RE.TA_log", main = "Histogram of RE.TA_log", col = "khaki1")
```



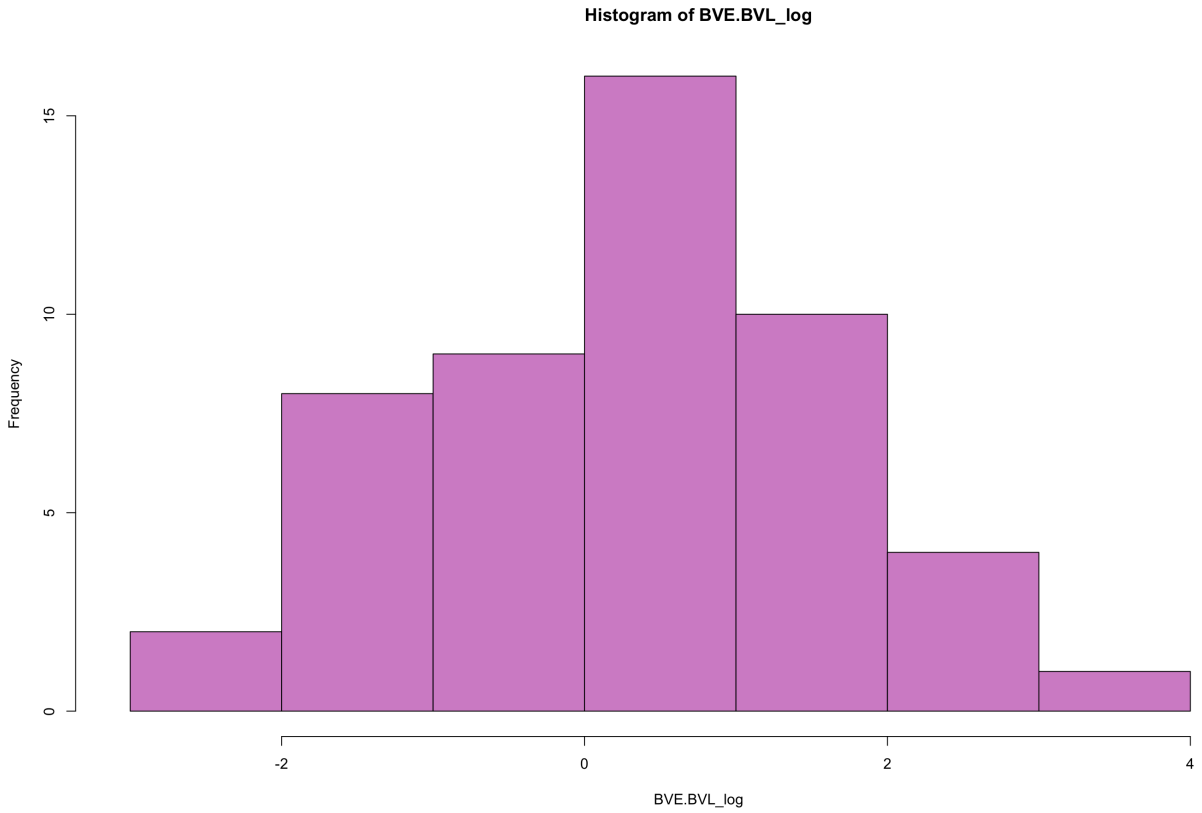
```r
#EBIT.TA_log
hist(EBIT.TA_log, xlim=range(min(EBIT.TA_log)-1, max(EBIT.TA_log)+1),xlab =
"EBIT.TA_log", main = "Histogram of EBIT.TA_log", col = "darkolivegreen1")
```




```r
#S.TA_log
hist(S.TA_log, xlim=range(min(S.TA_log)-1, max(S.TA_log)+1),xlab =
"S.TA_log", main = "Histogram of S.TA_log", col = "deepskyblue")
```




```r
#BVE.BVL_log
hist(BVE.BVL_log, xlim=range(min(BVE.BVL_log)-1, max(BVE.BVL_log)+1),xlab =
"BVE.BVL_log", main = "Histogram of BVE.BVL_log", col = "plum3")
```



The transformed indicators displayed a slight improvement in normality, making them more suitable for analysis. In particular, for EBIT.TA and BVE.BVL, the distributions are now closer to normal.

b) After coloring with red the bankrupted firms and in blue the non bankrupted firms, present all possible scatter plots that can be formed by the 5 covariates. Comment on which couple of variables jointly considered seem to have a good discriminatory power. Consider if some transformation (such as the log) of the data helps in achieving a better separation. Explore the command pairs (data) to build a matrix of all scatterplots.

```
par(mfrow=c(1,1))

color<-rep("blue",dim(dati)[1])
color[dati$Flag==0]<-"red"

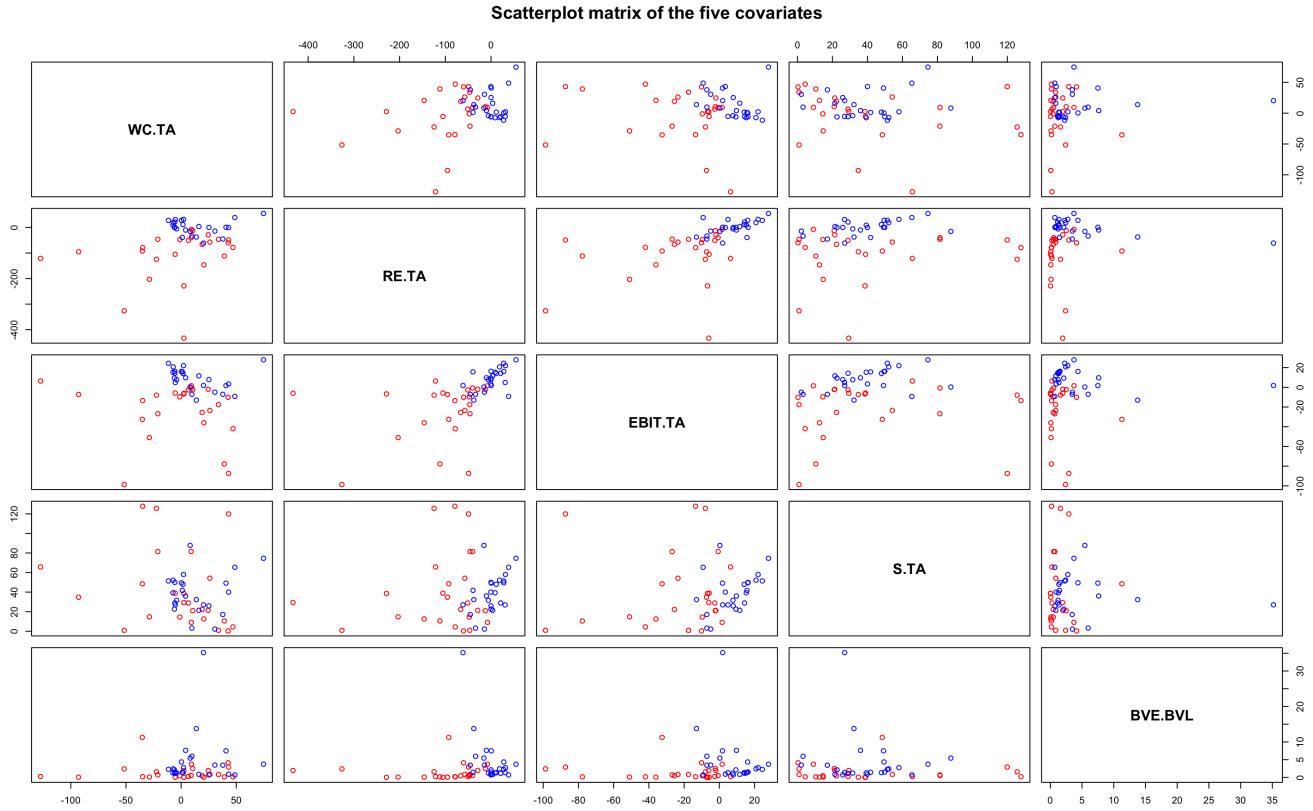
#WC.TA
plot(dati$WC.TA,dati$RE.TA, col=color,xlab = "WC.TA",ylab = "RE.TA",main =
"Scatterplot between WC.TA and RE.TA")
plot(dati$WC.TA,dati$EBIT.TA, col=color,xlab = "WC.TA",ylab = "EBIT.TA",main
= "Scatterplot between WC.TA and EBIT.TA")
plot(dati$WC.TA,dati$S.TA, col=color,xlab = "WC.TA",ylab = "S.TA",main =
"Scatterplot between WC.TA and S.TA")
plot(dati$WC.TA,dati$BVE.BVL, col=color,xlab = "WC.TA",ylab = "BVE.BVL",main
= "Scatterplot between WC.TA and BVE.BVL")
#RE.TA
```

```
plot(dati$RE.TA,dati$WC.TA, col=color,xlab = "RE.TA",ylab = "WC.TA",main =
"Scatterplot between RE.TA and WC.TA")
plot(dati$RE.TA,dati$EBIT.TA, col=color,xlab = "RE.TA",ylab = "EBIT.TA",main
= "Scatterplot between RE.TA and EBIT.TA")
plot(dati$RE.TA,dati$S.TA, col=color,xlab = "RE.TA",ylab = "S.TA",main =
"Scatterplot between RE.TA and S.TA")
plot(dati$RE.TA,dati$BVE.BVL, col=color,xlab = "RE.TA",ylab =
"BVE.BVL",,main = "Scatterplot between RE.TA and BVE.BVL")
#EBIT.TA
plot(dati$EBIT.TA,dati$WC.TA, col=color,xlab = "EBIT.TA",ylab = "WC.TA",main
= "Scatterplot between EBIT.TA and WC.TA")
plot(dati$EBIT.TA,dati$RE.TA, col=color,xlab = "EBIT.TA",ylab = "RE.TA",main
= "Scatterplot between EBIT.TA and RE.TA")
plot(dati$EBIT.TA,dati$S.TA, col=color,xlab = "EBIT.TA",ylab = "S.TA",main =
"Scatterplot between EBIT.TA and S.TA")
plot(dati$EBIT.TA,dati$BVE.BVL, col=color,xlab = "EBIT.TA",ylab =
"BVE.BVL",main = "Scatterplot between EBIT.TA and BVE.BVL")
#S.TA
plot(dati$S.TA,dati$WC.TA, col=color,xlab = "S.TA",ylab = "WC.TA",main =
"Scatterplot between S.TA and WC.TA")
plot(dati$S.TA,dati$RE.TA, col=color,xlab = "S.TA",ylab = "RE.TA",main =
"Scatterplot between S.TA and RE.TA")
plot(dati$S.TA,dati$EBIT.TA, col=color,xlab = "S.TA",ylab = "EBIT.TA",main =
"Scatterplot between S.TA and EBIT.TA")
plot(dati$S.TA,dati$BVE.BVL, col=color,xlab = "S.TA",ylab = "BVE.BVL",main =
"Scatterplot between S.TA and BVE.BVL")
#BVE.BVL
plot(dati$BVE.BVL,dati$WC.TA, col=color,xlab = "BVE.BVL",ylab = "WC.TA",main
= "Scatterplot between BVE.BVL and WC.TA")
plot(dati$BVE.BVL,dati$RE.TA, col=color,xlab = "BVE.BVL",ylab = "RE.TA",main
= "Scatterplot between BVE.BVL and RE.TA")
plot(dati$BVE.BVL,dati$EBIT.TA, col=color,xlab = "BVE.BVL",ylab =
"EBIT.TA",main = "Scatterplot between BVE.BVL and EBIT.TA")
plot(dati$BVE.BVL,dati$S.TA, col=color,xlab = "BVE.BVL",ylab = "S.TA",main =
"Scatterplot between BVE.BVL and S.TA")
```

We can represent all these combinations in a single scatterplot matrix.

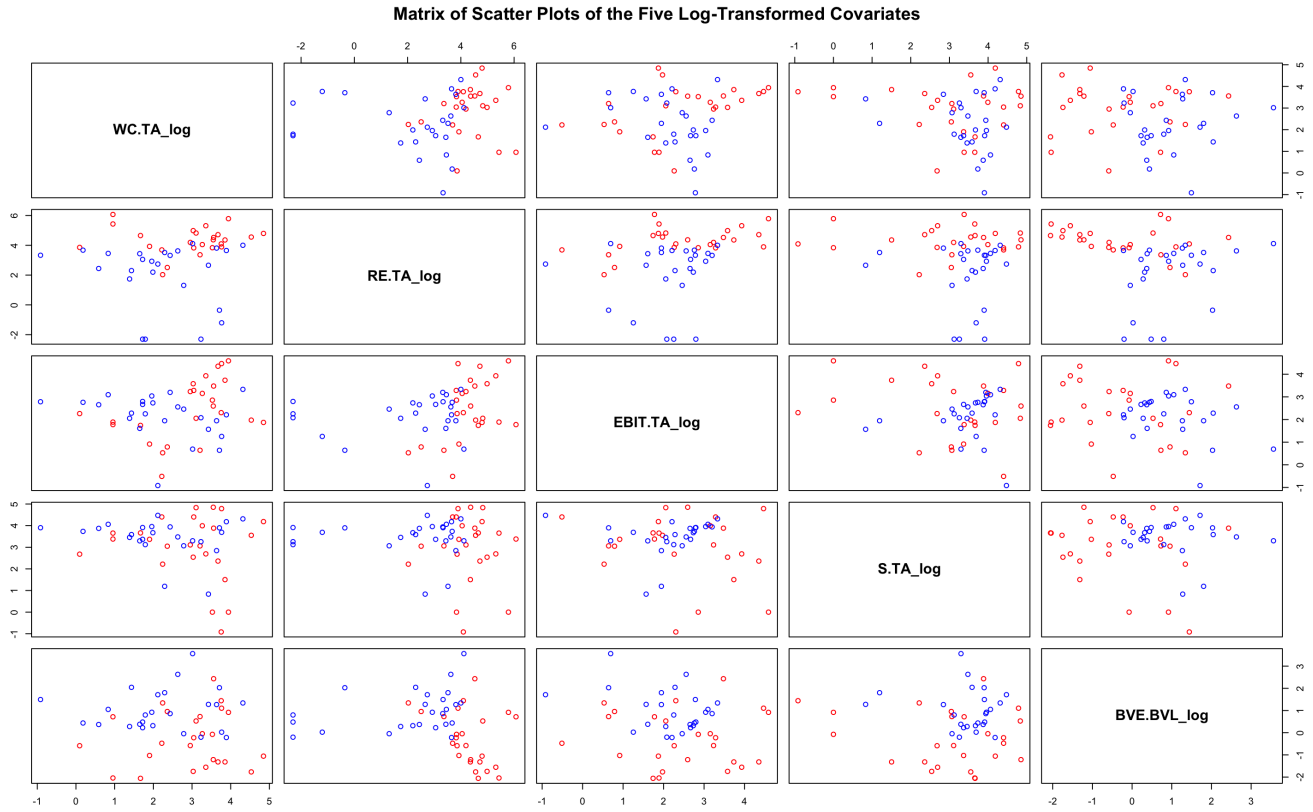
```
#we change the name of the variables to a better representation on the
matrix
WC.TA<-dati$WC.TA
RE.TA<-dati$RE.TA
EBIT.TA<-dati$EBIT.TA
S.TA<-dati$S.TA
BVE.BVL<-dati$BVE.BVL
```

```
#scatterplot matrix
pairs(~ WC.TA + RE.TA + EBIT.TA + S.TA + BVE.BVL,
      data = dati,
      col = color,
      main = "Scatterplot matrix of the five covariates",
      cex.labels = 1.5,
      font.labels = 2)
```



It might be useful to apply transformations to achieve a clearer separation.

```
pairs(~ WC.TA_log + RE.TA_log + EBIT.TA_log + S.TA_log+ BVE.BVL_log,
      data = dati,
      col=color,
      main = "Matrix of Scatter Plots of the Five Log-Transformed
Covariates",
      cex.labels = 1.5,
      font.labels = 2)
```



WC.TA and RE.TA are the best variables, compared to the others, which have very low discriminative power. This means that WC.TA and RE.TA show a clearer ability to differentiate between the groups, while the other variables are less effective in distinguishing them.

c) Let X be the r.v. representing log(BVE.BVL). In order to evaluate the discriminatory power of X (after possibly adding a small constant term, in order to make the log transform feasible), test the hypothesis that the expected value of the distribution of bads, μ0, is equal to the expected value of the distribution of goods, μ1, against the alternative hypothesis that they are different, i.e. H0 : μ0 = μ1 against the alternative H1 : μ0 < μ1, under the assumption that σ0 = σ1. Report p-value and discuss the implications in practical terms. Then redo the test by removing the assumption σ0 = σ1. Discuss the difference. Also, discuss which of the two tests is more suited for this data.

```
X<-BVE.BVL_log
#we create good and bad populations
P0<-X[dati$Flag == 0]  #Bankrupted firms (Flag = 0)
P1<-X[dati$Flag == 1]  #Not bankrupted firms (Flag = 1)
```

First, we created the r.v. X that represents log(BVE.BVL). We wanted to know the discriminatory power of X by doing a significance test where H0 : μ0 = μ1 against the alternative H1 : μ0 < μ1 (so testing if there isn't or not a difference between the expected value of the bad distribution and the good one); in two cases, first with σ0 = σ1 and then removing this assumption.

```r
#T TEST under the assumption σ₀ = σ₁
t_test_common_var <- t.test(P0, P1, alternative = "less", var.equal = TRUE)
```

```
        Two Sample t-test

data:  P0 and P1
t = -4.0767, df = 48, p-value = 8.545e-05
alternative hypothesis: true difference in means is less than 0
95 percent confidence interval:
       -Inf -0.7475738
sample estimates:
 mean of x  mean of y
-0.2765700  0.9935654
```

The first one is a Two Sample t-test, this test provides an unbiased estimate of the difference between the means. The p-value of 8.545e-05 is proof of strong evidence against the null hypothesis, meaning the difference between the groups is statistically significant. Moreover, the confidence interval (-Inf; -0.7475738) does not include zero, further confirming the statistical significance of the difference between the two groups.

```r
#T TEST under the assumption σ₀ ≠ σ₁
t_test_diff_var <- t.test(P0, P1, alternative = "less", var.equal = FALSE)
```

```
        Welch Two Sample t-test

data:  P0 and P1
t = -4.0767, df = 44.554, p-value = 9.285e-05
alternative hypothesis: true difference in means is less than 0
95 percent confidence interval:
       -Inf -0.7467773
sample estimates:
 mean of x  mean of y
-0.2765700  0.9935654
```

The significant test without the assumption (σ0 = σ1) is a Welch Two Sample t-test.
The p-value of 9.285e-05 supports the rejection of the null hypothesis and suggests a significant difference between the two means.

Also in this case the confidence interval does not include zero, confirming the statistical significance of the difference between means.

EXERCISE 2

```
dati=read.table(file.choose(),header=TRUE,comment.char="#")
```

a) Evaluate the relative frequency of "Good" clients among clients in the "North", and compare it with the "Center" and the "South". Interpret the results.

We obtain the table that divide the good and bad clients among the regions.

```
a<-table(dati$zres, dati$Flag)
```

```
          0    1
CENTRO  798  977
NORD    773 1089
SUD     803  836
```

```
freq_north_good<-a[2,2]/(a[2,2]+a[2,1]) #0.584855
# calculate observed probability of a unit that is in the Center to be good
freq_center_good<-a[1,2]/(a[1,1]+a[1,2]) #0.5504225
# calculate observed probability of a unit that is in the South to be good
freq_south_good<-a[3,2]/(a[3,1]+a[3,2]) #0.5100671
```

We can obtain the same results with the command prop.table:

```
prop.table(table(dati$zres, dati$Flag), margin = 1)
```

```
                0         1
CENTRO 0.4495775 0.5504225
NORD   0.4151450 0.5848550
SUD    0.4899329 0.5100671
```

Comparing the relative frequencies, in the Center 44.96% of clients are classified as bad, while 55.04% are good, meaning good customers slightly outnumber the bad ones. In the North, bad

customers make up 41.51%, while good ones account for 58.49%, making this the region with the highest percentage of good customers. In the South, the distribution is almost balanced, with 48.99% bad customers and 51.01% good ones. Overall, the North stands out for having the most good customers, while the South is the closest to an even split.

Evaluate now the relative frequency of "Good" clients that are in the "North" and compare with the same the Center and in the South. Interpret the results.

```
#good clients of each region over the sum of the good clients
#NORTH
freq_tot_north<-a[2,2]/(a[1,2]+a[2,2]+a[3,2]) #0.3752584
#CENTER
freq_tot_center<-a[1,2]/(a[1,2]+a[2,2]+a[3,2]) #0.3366644
#SOUTH
freq_tot_south<-a[3,2]/(a[1,2]+a[2,2]+a[3,2]) #0.2880772
```

The North has the largest share, with 37.53% of all good clients. The Center follows with 33.67%, still a significant portion but lower than the North. The South has the smallest share, with 28.81%, meaning it contributes the least to the total number of good clients.

Evaluate the relative frequency of clients who are in the "North" and are "Real Estate owner".

```
b<-table(dati$Flag, dati$zres, dati$imm)
```

```
, ,  = NO

   CENTRO NORD SUD
0     704  692 696
1     866  990 737

, ,  = SI

   CENTRO NORD SUD
0      94   81 107
1     111   99  99
```

```r
b1<-matrix(b[2,,],6,1) #GOOD clients: the first three rows aren't Real Estate Owners, instead the last three rows are Real Estate Owners
b2<-matrix(b[1,,],6,1) #BAD clients: the first three rows aren't Real Estate Owners, instead the last three rows are Real Estate Owners
B<-cbind(b2,b1) #we create a new table with bad clients in the column 1 and good clients in the column 2; not real estate owner in the first three rows and real estate owner in the last three rows; center in rows 1 and 4, north in rows 2 and 5, south in rows 3 and 6
```

```
     [,1] [,2]
[1,]  704  866
[2,]  692  990
[3,]  696  737
[4,]   94  111
[5,]   81   99
[6,]  107   99
```

```
freq_north_re<-(B[5,1]+B[5,2])/(B[5,1]+B[5,2]+B[2,1]+B[2,2]) #0.09667025
```

We computed the relative frequency of clients who are in the ”North” and are ”Real Estate owner” with the ratio between the sum of the clients in the North that are real estate owner and the sum of all the clients in the North.

Evaluate the relative frequency of the ”Good” clients among the total number of clients who are in the ”North” and are ”Real Estate owner”.

```
freq_good_north_re<-(B[5,2]/(B[5,1]+B[5,2])) #0.55
```

We computed the relative frequency of good clients among the total number of clients who are in the North and are Real Estate owner with the ratio between the good clients in the North that are real estate owner and the sum of good and bad clients in the North that are real estate owners.

b) Construct the 2×2 table of Flag against Imm. Let PY and PN be the probability a client, in order, who is a real estate owner and not real estate owner, to be good. Test the assumption H0 : PY = PN against the alternative H1 : PY≠ PN . Evaluate the p − value of the test and discuss results.

We create the table 2x2:

```
table_2x2 <- matrix(c(sum(B[4:6, 2]), sum(B[1:3, 2]),sum(B[4:6, 1]),
sum(B[1:3, 1])),nrow = 2, byrow = TRUE)
#names of the rows and columns
rownames(table_2x2) <- c("GOOD", "BAD")
colnames(table_2x2) <- c("Imm SI", "Imm NO")
```

	Imm SI	Imm NO
GOOD	309	2593
BAD	282	2092

```
PY<-table_2x2[1,1]/(table_2x2[1,1]+table_2x2[2,1]) #0.5228426
```

We obtain PY with the ratio between the good client that are real estate owner and all the clients that are real estate owner.

```r
PN<-table_2x2[1,2]/(table_2x2[1,2]+table_2x2[2,2]) #0.5534685
```

We obtain PN with the ratio between the good clients that aren't real estate owner and the sum of all the clients that aren't real estate owner.

PY<PN, so the clients that aren't real estate owner have a greater probability to be good.

Test the assumption H0 : PY = PN against the alternative H1 : PY ≠ PN . Evaluate the p − value of the test and discuss results.

```r
prop.test(c(table_2x2[1,1],
table_2x2[1,2]),c((table_2x2[1,1]+table_2x2[2,1]),
table_2x2[1,2]+table_2x2[2,2]),alternative = "two.sided",correct =
TRUE,conf.level = 0.95)
```

```
        2-sample test for equality of proportions with continuity correction

data:  c(table_2x2[1, 1], table_2x2[1, 2]) out of c((table_2x2[1, 1] + table_2x2[2, 1]),
table_2x2[1, 2] + table_2x2[2, 2])
X-squared = 1.867, df = 1, p-value = 0.1718
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.07428968  0.01303792
sample estimates:
   prop 1    prop 2
0.5228426 0.5534685
```

The p-value is quite high (> 0.05), so we do not reject the null hypothesis. This suggests that there is no statistically significant difference between the two proportions. Also the confidence interval confirms the result obtained with the p-value, indeed it doesn't contain the zero.

c) From the theoretical point of view, which of the tests performed at points c) of Exercise 1 and b) of Exercise 2 is in the logic of the discriminant analysis and which is in the logic of the logistic regression? Explain why.

From a theoretical point of view, the t-test analyzed in point c of Exercise 1 was used in the logic of discriminant analysis. The goal of this type of analysis is to examine the separation between groups based on continuous variables, assuming that the variables follow a specific distribution (e.g., a normal distribution in the case of the common variance model). The test was conducted to compare the means of the two population groups to determine whether they differ significantly.
On the other hand, the test performed in point b of Exercise 2 follows the logic of logistic regression, as it analyzes differences in category proportions. In this case, the test aims to

compare the probabilities of belonging to a specific category to assess whether the two examined probabilities (PY and PN) differ significantly.

d) Randomly divide the whole sample into a validation sample of 500 observations and a training sample of 4776. Do this by randomly selecting 500 observations (use the function a < −sample.int(n1, n2, replace = FALSE) which extracts n2 integers numbers from 1 to n1 without replacement), then use validazione < −data[a,] and sviluppo < −data[−a,].

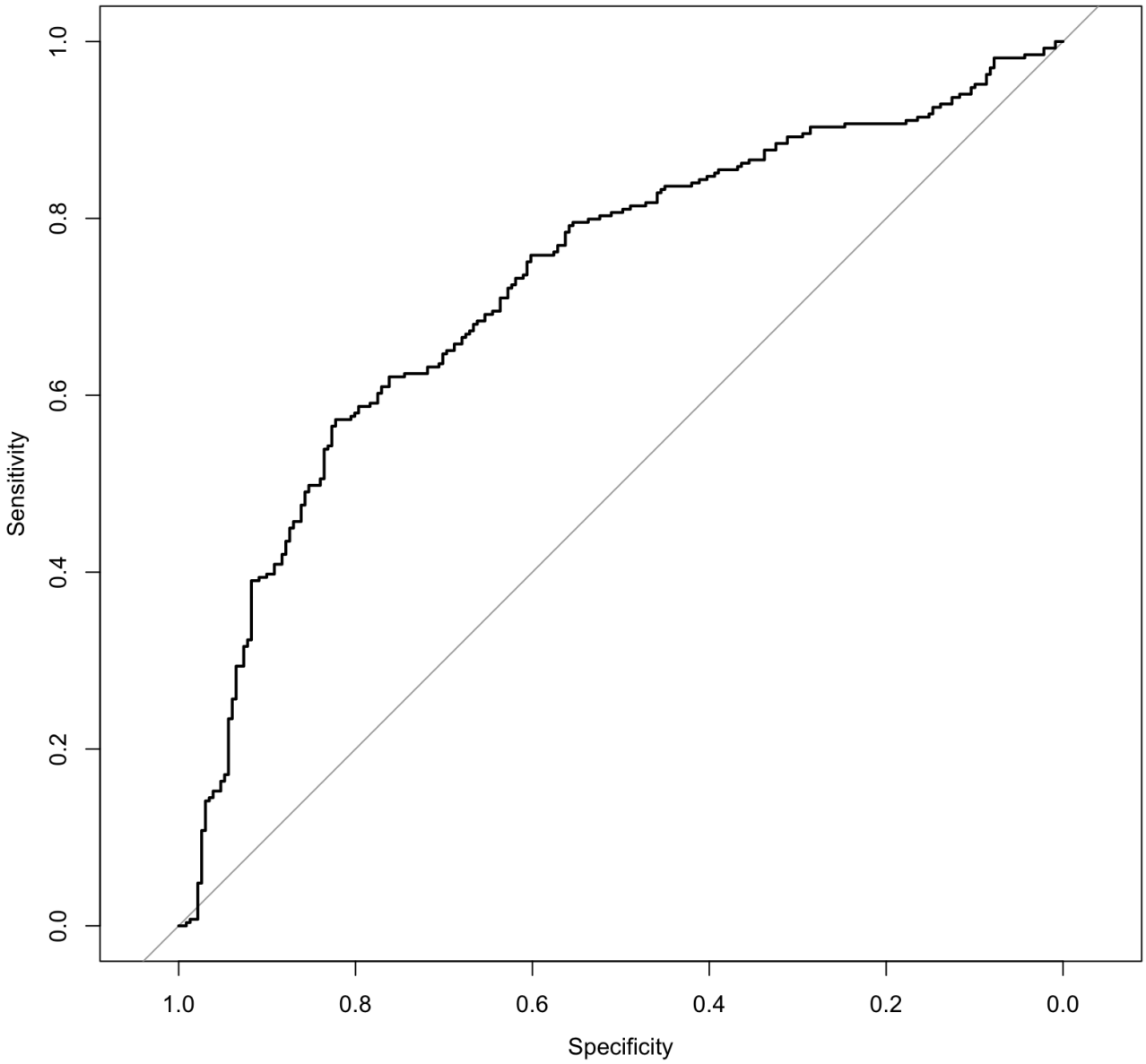
```
set.seed(123) #we fix the seed
s <- sample.int(nrow(dati), 500, replace = FALSE)

validazione <- dati[s, ]
sviluppo <- dati[-s, ]
```

e) By ignoring the binary nature of the Flag, estimate on the training sample the following linear regression function Flag = a + b1lastsal + b2kept + ε. Use the estimated coefficients to build your scoring function sˆ(x) = ˆa + ˆb1lastsal + ˆb2kept. Evaluate the ROC curve of the scoring funtion on the validation sample (N.B. you may use the command predict). Import the package pROC with the command library(pROC) and use the command roc(validazione§flag,score.val, plot=TRUE) where score.val is the score function built on the validation sample. Comment.

```
mod<-lm(Flag~lastsal+kept, data=sviluppo) #create the model
summary(mod)
coef(mod) #the score function is s_hat(x)=0.227-0.000023*lastsal+0.6705*kept
score.val <- predict(mod, newdata = validazione)  #score.val is the score
function
```

```
library(pROC)
par(mfrow=c(1,1))
roc(validazione$Flag, score.val, plot=TRUE)
```




```
Call:
roc.default(response = validazione$Flag, predictor = score.val,     plot = TRUE)

Data: score.val in 231 controls (validazione$Flag 0) < 269 cases (validazione$Flag 1).
Area under the curve: 0.7279
```

The range for the Area Under the Curve (AUC) is 0.0-1.0. Our model has an AUC of 0.7279, so it performs quite well, but it is not excellent.

f) (Theoretical) If we used as a cut-off the first quartile of the distribution of bads, how much is the probability of a type I error? If, instead, we used the third quartile of the distribution of goods, how much is the probability of type I error?

If we set the first quartile (Q1) of the "Bad" distribution as the cut-off, we classify a customer as "Good" only if their score is higher than Q1.

Using this cut-off usually results in a low Type I error probability. This happens because a client is labeled as "Good" only if their score ranks among the top 25% of the "Bad" customers. Since Q1 is a relatively low threshold, more customers will be classified as "Good," but the chance of misclassifying a "Good" client as "Bad" remains relatively low.

That said, even with a lower cut-off, the probability of a Type I error is never zero—just lower than it would be with a higher cut-off.

On the other hand, if we use the third quartile (Q3) of the "Good" distribution as the cut-off, a client is classified as "Good" only if their score is higher than Q3.

Using Q3 as a cut-off increases the probability of a Type I error. This happens because a client is considered "Good" only if their score falls within the top 75% of the "Good" distribution. While this approach makes the classification more selective, it also raises the likelihood of incorrectly labeling "Good" customers as "Bad."

In this case, the risk of misclassification is higher than with the Q1 cut-off, meaning the Type I error probability is greater.

EXERCISE 3

From a priori information we know that the ratio between the two costs C(1|0)/C(0|1) is 1.2, while the a priori probability P(Y = 1) that a SME is good is 0.75. Let X be the r.v. that represents a balance sheet indicator. We know that the conditional distribution f1(x) is Gaussian with expected value μ1 = 63 and σ2 1 = 40, while f0(x) is Gaussian with expected value μ0 = 39 e σ2 0 = 30. Next applicant has a value of X of 52.

To solve this exercise, the first step is to write down the data that will be useful for computing the various elements necessary for the decision making approach and customer classification.

```
#Below, we define the given parameters:
mu1 <- 63  # Mean for class Y = 1
var1 <- 40 # Variance for class Y = 1
mu0 <- 39  # Mean for class Y = 0
var0 <- 30 # Variance for class Y = 0
x <- 52 # Observed value of X
P_Y1 <- 0.75 # Prior probability P(Y = 1)
P_Y0 <- 0.25 # Prior probability P(Y = 0)
cutoff <- 1.2 # Ratio between costs
```

a) By using the criterion that minimises the expected loss, decide whether to grant or not the new applicant.

To do this we have to follow different steps: the first one is to compute the conditional density functions f1(x) and f0(x) given the values of Y and evaluated in x.

```
f1_x <- dnorm(x, mean = mu1, sd = sqrt(var1)) # Likelihood for Y = 1
f0_x <- dnorm(x, mean = mu0, sd = sqrt(var0)) # Likelihood for Y = 0
```

The second step is to find the posteriori probabilities P(Y=1|x) and P(Y=0|x) using also the density functions f1(x) and f0(x) computed previously:

```
P_Y1_x <- (f1_x * P_Y1) / (f1_x * P_Y1 + f0_x * P_Y0) # P(Y=1|x)
P_Y0_x <- (f0_x * P_Y0) / (f1_x * P_Y1 + f0_x * P_Y0) # P(Y=0|x)
```

These posterior probabilities are necessary because, in order to define the acceptance region, also called A1, within which the customer is classified as good, the following rule must be satisfied: the ratio of the posterior probabilities P(Y=1|x)/P(Y=0|x) must be greater than a cut-off score, also said c.
The optimal cut-off score is the ratio between the cost C(0/1) that must be paid if a unit from P1 is assigned to P0 (cost of type I error) and viceversa C(1/0) (cost of type II error).
If we compute the ratio between posterior probabilities:

```
post_prob_ratio <- P_Y1_x/P_Y0_x
```

We can notice that the ratio of the two posterior probabilities (9.572917) is greater than the ratio of the two costs (1.2), therefore we will proceed to accept the customer.

```
#Decision:
decision <- if(post_prob_ratio > cutoff) "Grant the new applicant" else "Not
grant the new applicant"
```

b) Evaluate the posterior probability P(Y = 1 | x) that such applicant is a good client.

```
P_Y1_x <- (f1_x * P_Y1) / (f1_x * P_Y1 + f0_x * P_Y0) # P(Y=1|x)
```

Probability already calculated In point (a) that consider the flag = 1 (SME is a good firm).

c) Suppose that we choose 60 as a cut-off, i.e. we accept all applicants with X ≥ 60. Evaluate the probability of type I error, of type II error and the corresponding x-value and y-value on the ROC curve.

```r
#Create a new variable for the new cutoff point
cutoffvalue <- 60
#Compute Type I Error (we reject a good unit, so, we consider a good unit
like a bad one)
type_I_error <- pnorm(cutoffvalue, mean = mu1, sd = sqrt(var1), lower.tail =
TRUE)
#Compute Type II Error(we do not reject a bad unit and we consider a bad
unit like a good one)
type_II_error <- pnorm(cutoffvalue, mean = mu0, sd = sqrt(var0), lower.tail
= FALSE)
```

The ROC is the curve on the cartesian plan that matches the points with x-value 1-F (s|Y=0) and y-value 1−F (s|Y=1).

```r
#x-value and y-value on the ROC curve
x_roc <- type_II_error #false positive rate also known as specificity
y_roc <-1 - type_I_error #true positive rate also known as sensitivity
```